Tanishq Annavaram

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Education

Rutgers University - New Brunswick, NJ — Dec 2026
Bachelor of Science in Computer Science — ***3.75*** *GPA*
Relevant Coursework: Computer Architecture, Intro to **AI**, Database Management Systems**(DBMS)**, Data Science, Algorithms

Technical Skills

Languages: Python, R, C/C++, SQL, Java, JavaScript, HTML/CSS, Kotlin, PHP, XML,
Frameworks: PyTorch, TensorFlow, Node.js, Express, React
Databases: MongoDB, MSSQL, MySQL, Redis, PostgreSQL
Libraries: Pandas, NumPy, Matplotlib, Scipy, Scikit-learn, Streamlit

Projects

Distributed Event-Driven Order Processing System | *Python, FastAPI, PostgreSQL, Redis, RabbitMQ, Docker, GitHub Actions*

- Built and deployed a containerized event-driven order platform using RabbitMQ, PostgreSQL, Redis, Docker Compose, and GitHub Actions CI/CD.
- Improved reliability with transactional outbox, idempotency keys, retries, dead-letter queues, and reconciliation jobs for distributed workflow recovery.
- Added OpenTelemetry, Prometheus, and Grafana dashboards, and validated cross-service flows with Testcontainers against real infrastructure dependencies.

Splitshare-Expense Splitting Tool | *JavaScript, Node.js, Express.js, React.js, MongoDB, REST API design*

- Engineered a full-stack expense-sharing application (MERN stack) that automated group bill settlements, reducing manual calculation time by 80%.
- Optimized debt resolution using a greedy O(n log n) algorithm, decreasing required transactions by 50–70% in test groups of 10+ users.
- Built and documented RESTful APIs in Express.js with MongoDB, improving response efficiency by 40%.

Experience

Software Engineer Intern — Apr. 2025 – Aug. 2025
BizBites Technologies — *Remote*

- Built a secure backend with Node.js, Express.js, and MySQL, implementing bcrypt-based authentication that reduced login errors by 40%.
- Developed RESTful APIs for device/switch management with transactional queries, improving database performance by 35% and cutting device response latency by 25%.
- Streamlined deployment using Docker, Postman, and GitHub Actions (CI/CD), accelerating release cycles by 50%.

Data & Machine Learning Intern — Mar. 2026 – Present
Algorithmic Control and Robotics Lab — *New Brunswick, NJ*

- Developed a transformer-based next-state prediction framework for robotic manipulation, extending prior DIPN work on multi-object push interaction modeling.
- Built object-centric learning pipelines using RGB-D inputs, simulation data, and structured object representations to model how pushes propagate across multiple objects in clutter.
- Supporting experimental analysis through attention-map visualization, trajectory analysis, and IoU / manipulation-efficiency evaluation within a broader robotic manipulation pipeline

Research Assistant — Aug. 2025 – Present
Center of Advanced Infrastructure and Transportation(CAIT), Rutgers University — *New Brunswick, NJ*

- Supported a YOLO-based computer vision pipeline for railroad trespassing detection through video preprocessing, annotation, validation, and QA on multi-site real-world footage.
- Performed error analysis on model outputs and converting raw detections into structured datasets and short evidence clips for downstream analysis
- Developed clear technical documentation and data guides for research teams and non-tech staff, improving workflow consistency and reducing clarification requests by 30%.